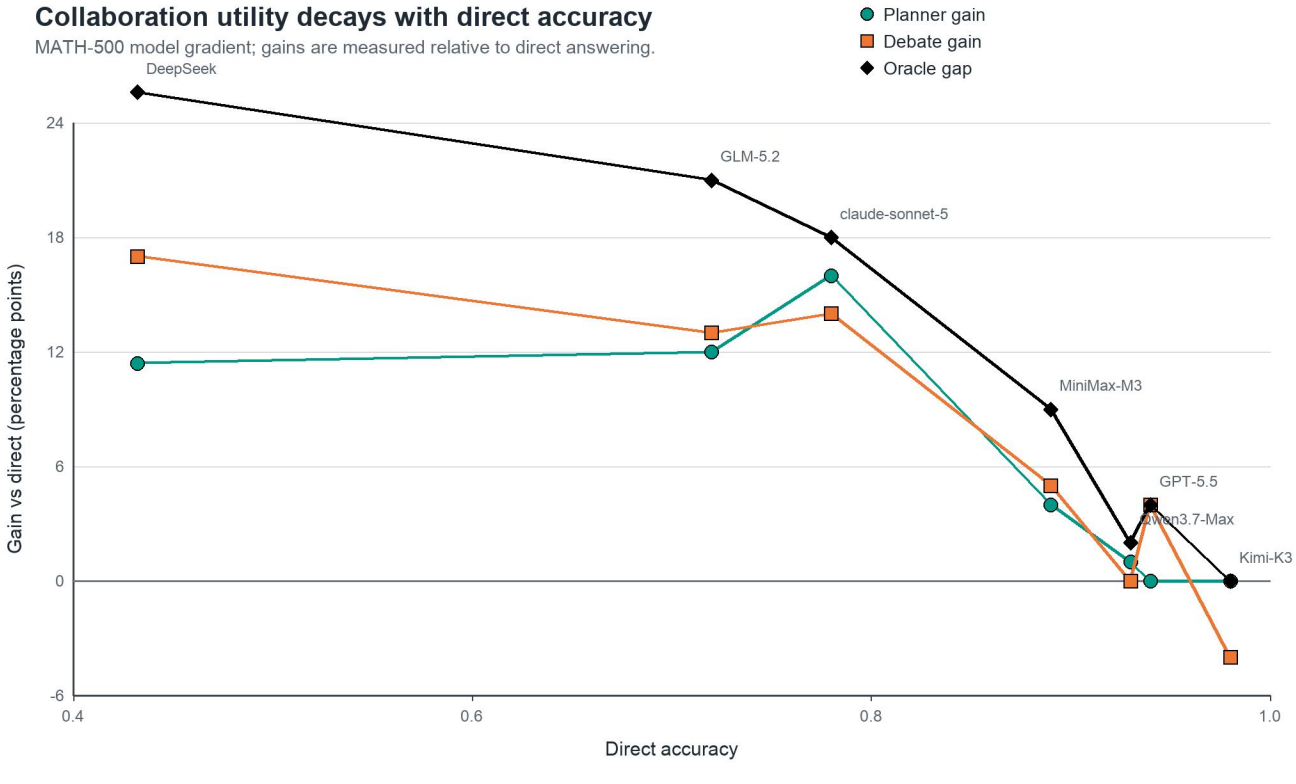

Collaboration utility decays with direct accuracy
MATH-500 model gradient; gains are measured relative to direct answering.
Planner gain
Debate gain
Oracle gap
DeepSeek
GLM-5.2
claude-sonnet-5
MiniMax-M3
GPT-5.5
Qwen3.7-Max
Kimi-K3
Gain vs direct (percentage points)
Direct accuracy
24
18
12
6
0
-6
0.4
0.6
0.8
1.0